How to Kernelize SVM

Example: ***Quadratic Kernel***

$$\mathcal{L}_D(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{j=1}^{N} \sum_{i=1}^{N} \alpha_i \alpha_j y_i y_j$$ [ ]

$$W = \sum_{} \alpha_i y_i X_i$$

$$b = -\frac{1}{2}\{ \max_{i,\, y=-1} [\ ] + \min_{i,\, y=+1} [\ ] \}$$

Class of vector **Z** ⇒ sign(W^T Z [ ] + b)

Kernelize using

$$\mathbf{K}(\alpha, b) = \alpha^T b^2$$

$$\mathcal{L}_D(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{j=1}^{N} \sum_{i=1}^{N} \alpha_i \alpha_j y_i y_j$$ [ ]

$$W = \sum_{} \alpha_i y_i X_i$$

$$b = -\frac{1}{2}\{ \max_{i,\, y=-1} [\ ] + \min_{i,\, y=+1} [\ ] \}$$

Class of vector **Z** is sign($[W^T Z]^2$ [ ] + b)